

# Module 01: Process Concepts, Process States & PCB

## 1. Process Concept: Program vs Process

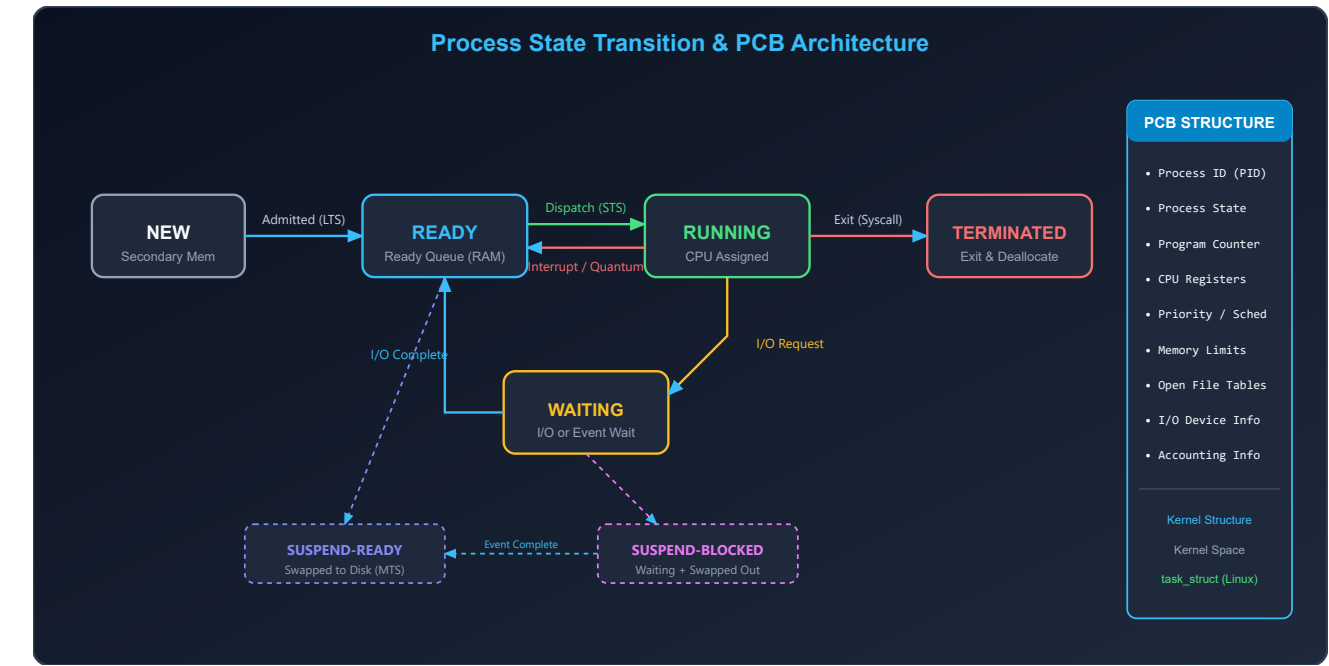
**Program (Passive):** Disk par stored static machine binary code file hai.

**Process (Active):** Program in execution. Iske paas memory address space, allocated CPU registers, stack, aur heap hota hai.

### Process Address Space Layout

| Memory Segment | Description                                                            | Growth Direction               |
|----------------|------------------------------------------------------------------------|--------------------------------|
| Stack          | Local variables, function return addresses, parameters                 | High to Low Memory (Downwards) |
| Heap           | Dynamic runtime allocation (malloc, calloc, new)                       | Low to High Memory (Upwards)   |
| Data           | Global and static variables (Initialized .data and Uninitialized .bss) | Fixed at launch                |
| Text           | Compiled read-only executable instructions                             | Fixed                          |

## 2. Process State Transition Model



Key Transition Rules:

1. New → Ready:

Long-Term Scheduler (LTS) dwara admission.

2. Ready → Running:

Short-Term Scheduler (STS) / Dispatcher allocation.

3. Running → Ready:

Timer interrupt (Preemption) ya higher priority process arrival.

- 4. **Running** → **Waiting**: I/O request ya event wait system call.
- 5. **Waiting** → **Ready**: I/O completion interrupt (Directly running nahi ja sakta!).

### 3. Process Control Block (PCB) Fields

| PCB Field        | Purpose in OS Kernel                                       |
|------------------|------------------------------------------------------------|
| PID / PPID       | Unique process identifier and Parent process ID            |
| Process State    | Ready, Running, Waiting, Terminated                        |
| Program Counter  | Next instruction address to be executed on CPU resume      |
| CPU Registers    | Saved hardware registers context during preemption         |
| Memory Limits    | Base/Limit registers or Page Directory base register (CR3) |
| File Descriptors | Array of open file handles and I/O sockets                 |